# And the winner is…

## How to make compromises between illocution, information structure and prosodic requirements for accent placement

Heiko Seeliger

Universität zu Köln

Linguistischer Arbeitskreis Köln
2021-12-08

# This talk in a nutshell

- Topic of this talk: Prosodic marking of information structure in questions, particularly the marking of givenness
- Since questions are used to pursue entirely different communicative goals than assertions, this can shed light on what can (and can't) be done with language
- Main takeaways: (i) alternatives (in the focus sense) can play different roles in questions than in assertions, (ii) givenness-marking in questions is not done for the benefit of the addressee

- The data presented here is from joint work with Sophie Repp as part of Project A06 of the CRC 1252 "Prominence in Language".

# Outline

1. Givenness & contrast
2. Two case studies on accent placement in questions
3. Double accents in questions and assertions

# Outline

1. Givenness & contrast
2. Two case studies on accent placement in questions
3. Double accents in questions and assertions

# A note on notation

- Accents are marked in CAPITALS.
- Deaccented words are not specifically marked.

(1) A: Who arrested Smith?

B: The DETECTIVE arrested Smith.[1]

- In cases of double accents (*The DETECTIVE arrested SMITH*), the notation is agnostic on the identity of nuclear, pre-nuclear and post-nuclear accents.

[1] (1) from Wagner (2012)

# Default rules for accent placement

- In German, the default nuclear accent is realized on the most deeply embedded, non-pronominal argument, i.e. in transitive sentences on the direct object:

(2) Der Detektiv verhaftet SCHMIDT.
*the detective arrests Schmidt*
'The detective arrests SCHMIDT.'

- Focus, contrast and givenness can shift this accent to other words, deriving marked accentuation patterns.
- A note specifically on German: the underlying default accent pattern is "OBJECT verb", since the German VP is right-headed:

(3) Der Detektiv hat SCHMIDT verhaftet.
*the detective has Schmidt arrested*
'The detective has arrested SCHMIDT.'

# Givenness

- As a baseline: what do we know about prosodic givenness marking in assertions?
- If a word refers to given information, it can be deaccented (even if it is the location of the default nuclear accent).
- For various reasons, deaccentuation is better referred to as accent shift.
- Givenness-induced accent shift in assertions can be:
  - mandatory
  - optional
  - impossible

# Mandatory accent shift

(4) A: Smith walked into a store. What happened next?

a. #B: A detective arrested SMITH.

b. B: A detective ARRESTED Smith.[2]

- (4a), if it can be said at all, presupposes the existence of a second person also named *Smith* (and is too vague as a consequence).

[2] (3) from Wagner (2012)

# Optional accent shift

(5) A: John's aunt, who is incredibly rich and owns a bicycle factory, came to his wedding. I wonder what she brought as a present.

a. B: Guess what: she brought a used BICYCLE.

b. B: Guess what: she brought a USED bicycle.[3]

- There is a contextual expectation that if John's aunt brings a bicycle, it will be a new one (since she owns a bicycle factory).
- The implicit contrast between *new* and *used* can motivate the accent shift in (5b).

[3] (36) from Wagner (2012)

# Impossible accent shift

(6) A: Sally's uncle, who is incredibly rich and produces expensive/luxury convertibles, came to her wedding. I wonder what he brought as a present.

a. B: Guess what: he brought an expensive CONVERTIBLE.

b. ?#B: Guess what: he brought an EXPENSIVE convertible.[4]

- In contexts in which everything is given, it is usually not possible to shift the accent from its default location.

[4] (38) from Wagner (2012)

# Contrast

- It has been claimed that prosodic givenness marking depends on the presence of mutually exclusive alternatives, i.e. on focus (e.g. Wagner, 2012; Büring, 2019).
- These alternatives can be implicit, i.e. non-overt.
- We will also need a term for focus-marked words in the overt presence of alternatives (cf. Repp, 2016 on degrees of contrast).

→ Contrastive focus

# Outline

1. Givenness & contrast
2. Two case studies on accent placement in questions
3. Double accents in questions and assertions

# General information on experimental design

- Data from two studies: Seeliger and Repp (submitted) and Seeliger and Repp (in preparation)
- Both were production studies in which participants listened to and read parts of pseudo-dialogues.
- We manipulated information structure on two separate axes:
  - contrast vs. lack of contrast:
    explicit alternative in the context?
  - given vs. new information:
    previous mention of referring expression?

# Experiment I: target sentences

- We elicited polar questions. They have the following characteristics:
  - Syntax: verb-first
  - Used to elicit a yes/no answer from the addressee
  - Embedded between two other sentences that clarify speaker intent

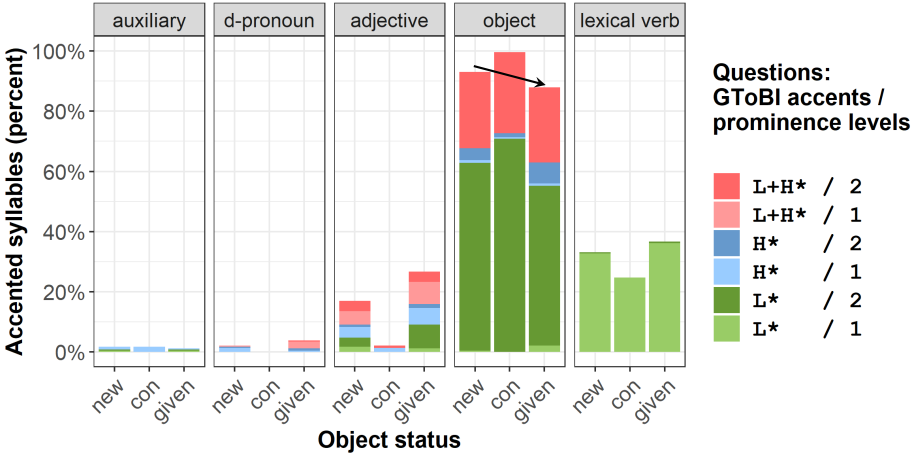
# Experiment I: materials

(7) A: Have you heard about the results of the research trip to South America? They went to the Andes to look for particularly small species of {NEW: **domestic animals** / GIVEN: **llamas**} that hadn't been described previously. And they were really successful. Frank, for instance. He was really lucky.

B: Ach echt? **Hat der kleine Lamas gefunden?** Das würde ich spannend finden.
*Oh really? Did he find small llamas? I would find that interesting.*

(There was also a contrastive condition, which I'll skip today.)

# Experiment I: results




- No (statistically significant) deaccentuation of given objects in these questions!

# Experiment I: comparison with fillers

- As a sanity check, we wanted to know whether the participants *ever* deaccented given material.
- Conveniently, the fillers also contained given objects.
- The fillers consisted of declarative questions (8a) and rejections (8b).

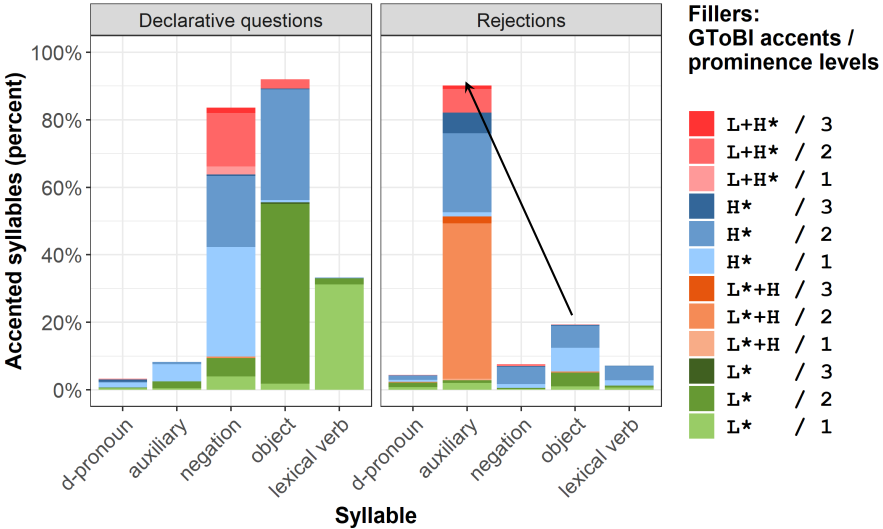
(8) *[Context for b:]* A: I loved Julia's idea that all her guests bring flowers instead of presents. It smells so nice in here! I'm especially looking forward to what Torsten brought. As you know, he works in a plant nursery.

a. B: Der hat keine Blumen mitgebracht?

b. B: Der hat keine Blumen mitgebracht!

*He didn't bring any flowers?/!*

# Experiment I: filler results



# Lack of deaccentuation: analysis

(9) a. Hat der kleine LAMAS gefunden? [Default]

b. Hat der KLEINE Lamas gefunden?

c. HAT der kleine Lamas gefunden?

d. Hat der kleine Lamas GEFUNDEN?

- Each of the non-default accent placements in (9) can (and does) have semantic effects.
- Semantic effects must be compatible with the context.
- In the material, all other referring expressions were also either given (d-pronoun, adjective) or accessible (lexical verb).

→ Accent shift from given material to other material is blocked – would indicate narrow focus, which isn't contextually licensed.

# Lack of deaccentuation: analysis

- What about the finite auxiliary verb? It's not given in a meaningful sense.
- Accent on the auxiliary would indicate VERUM focus (Höhle, 1992).
- VERUM in questions indicates either
  - epistemic bias – i.e. the speaker of the question had assumed that Frank had NOT found any small llamas.
  - impatience on the part of the speaker (Biezma, 2009).
- Neither of those two effects is compatible with the very accepting context of the questions (*Oh really?* […] *I would find that interesting.*)
- VERUM focus *is* licensed in rejections by focus on the polarity of the proposition → accent shift does happen in rejections

# Experiment I: summary

- Givenness is necessary, but not sufficient to license accent shift.
- In all-given questions, the nuclear accent tends to be realized in its default location (if VERUM focus is blocked).
- This parallels assertions, as we have seen:

(6) A: Sally's uncle, who is incredibly rich and produces expensive convertibles, came to her wedding. I wonder what he brought as a present.

a. B: Guess what: he brought an expensive CONVERTIBLE.

b. ?#B: Guess what: he brought an EXPENSIVE convertible.

# Experiment II: introduction

- Similar paradigm as Experiment I (production study, read speech, pseudo-dialogues)
- Verb-final *wh*-questions instead of polar questions
  - Note: these are embedded questions, and the embedding structure was a polar question.
  - However, the results should generalize to matrix *wh*-questions and simple polar questions.
- Subject and object are always given; lexical verb is accessible.
- Experimental manipulation: contrastive alternative to the subject of the sentence present/absent in the context

# Experiment II: materials

No contrastive alternative for the subject:

(10) A: Have you already heard? Anna has now focused on Germanic tribes for her dissertation.

B: Wirklich? Dann ist sie bestimmt viel unterwegs, um an Originalquellen von Germanen heranzukommen. Weißt du zufällig, **wo die schon überall Germanen erforscht hat?**

*Really? Then she's probably traveling a lot to obtain original sources on Germanic tribes. Do you happen to know* ***where she has already researched Germanic tribes?***

# Experiment II: materials

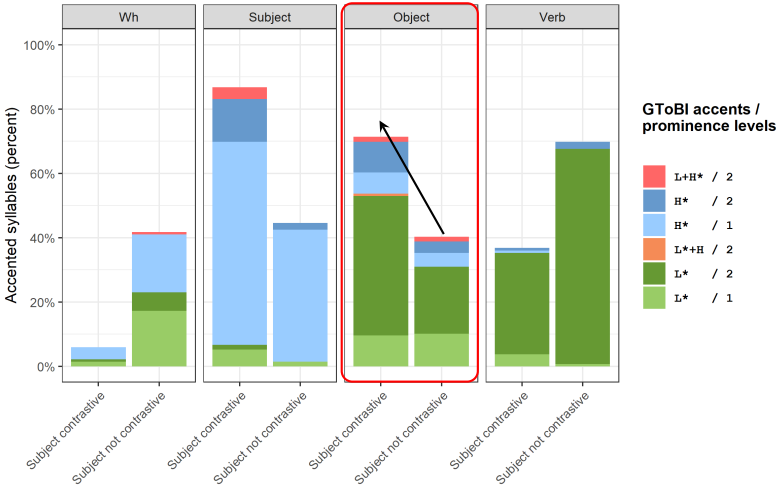
Contrastive alternative for the subject:

(11) A: Have you already heard? Paul has now focused on Germanic tribes for his dissertation.

B: Wirklich? Da ist er bestimmt viel unterwegs, um an Originalquellen von Germanen heranzukommen. Aber ich könnte mir vorstellen, dass das noch nichts gegen die Reisen von Anna ist. Hast du zufällig mal mitbekommen, **wo die schon überall Germanen erforscht hat**?

*Really? Then he's probably traveling a lot to obtain original sources on Germanic tribes. But I can imagine that that's nothing compared to the travels of Anna. Do you happen to have heard* ***where she has already researched Germanic tribes?***

# Experiment II: results




- Contrast on the **subject** leads to an increased number of **object** accents!

# Accounting for the contrast findings I

(12) But I can imagine that that's nothing compared to the travels of Anna. …

B: Where has SHE already researched Germanic tribes?[5]

B': Where has SHE already researched GERMANIC tribes?

- After the contrast to Anna is made, we only know that Anna also goes on research trips, and that they are probably more extensive/numerous than Paul's research trips.
- We don't know whether Anna also researches Germanic tribes, or another old European tribe, or something else entirely.

[5] In the following, I will discuss **matrix** questions instead of embedded ones.

# Accounting for the contrast findings II

(12) But I can imagine that that's nothing compared to the travels of Anna. …

B: Where has SHE already researched Germanic tribes?

B': Where has SHE already researched GERMANIC tribes?

- If the speaker assumes that Anna also researches Germanic tribes, and wants to convey this assumption, they should deaccent the object (12B).
- If the speaker does NOT want to convey that assumption, they should accent the object (12B').
- The givenness marking is truly optional, depending on the discourse commitments of the speaker.

# Accounting for the contrast findings III

(13) a. <u>Unaccented object:</u>
QUD 1: What about ANNA's research on Germanic tribes?
QUD 1.1: Where has SHE already researched them?

b. <u>Double accent:</u>
QUD 2: What about ANNA's research?
QUD 2.1: In particular, what about research on GERMANIC TRIBES?
QUD 2.1.1: Where has SHE already researched THOSE?

- Crucially, the accent on the object does not appear to be contrastive.
- None of the focus alternatives to *Germanen* would (automatically) be excluded. (Where "exclusion" needs to mean something like "would make the answer to the question be the empty set".)

# Optional givenness marking

- Another perspective: givenness marking of the object does not automatically ‘carry over’ across the contrastive topic change.
- The speaker is free to treat the object as given – or not.
- This seems to have no bearing on the answer possibilities offered to the hearer.
- Let’s take a more in-depth look at the discourse effects of double accent structures, both in questions and in assertions.

# Outline

1. Givenness & contrast
2. Two case studies on accent placement in questions
3. Double accents in questions and assertions

# Double focus in assertions

(14) John cited Mary, but …

a. …he DISSED SUE.

b. ? …he dissed SUE.[6]

- Double contrast – *Mary* vs. *Sue*; *cite* vs. *diss* (note: both contrasts are explicit.)

(15) Fred read the menu and then …

a. …he ordered a HAMburger.

b. # …he ORdered a HAMburger.[7]

- No contrasts – *read* vs. *order* or *menu* vs. *hamburger* → no double accent possible

[6] (60) from Schwarzschild (1999)

[7] (35) from Büring (2019)

# Double accents with only one explicit contrast

- So two explicit contrasts can lead to a double focus structure in assertions.
- What about one explicit contrast? Like this:

(12) *[Context about Paul]* …But I can imagine that that's nothing compared to the travels of Anna. …

B: Where has SHE already researched GERMANIC TRIBES?

B': Where has SHE already researched Germanic tribes?

- Do we also find accent patterns like (12B) in assertions?
- First off, we need to make sure we're not comparing apples and oranges.

# Preliminaries: speech act differences

- Assertions and questions differ fundamentally in the fact that the speaker of an assertion knows about true states of affairs. The speaker of a question aims to find out what is true.

(5) A: John's aunt, who is incredibly rich and owns a bicycle factory, came to his wedding. I wonder what she brought as a present.

a. B: She brought a used BICYCLE.

b. B: She brought a USED bicycle.

- *used* can be accented because it contrasts with the expectation that the aunt would bring a *new* bike.
- *used* can be asserted in the first place because the state of affairs is such that the proposition is true – which B knows.

# Preliminaries: speech act differences

(16) A: John's aunt, who is incredibly rich and owns a bicycle factory, came to his wedding. Guess what she brought as a present.

a. #B: Did she bring a used BICYCLE?

b. ??B: Did she bring a USED bicycle?

- Nothing makes *used* salient in the question context. B does not know whether the actual state of affairs is such that *used* is relevant.
- (Another perspective: the question's evidential bias (cf. Sudo, 2013) is not satisfied.)
- (16b) might be improved with a continuation like *You know, since she's a cheapskate*. Unlike with assertions, this accommodation must be mentioned explicitly.

# Double focus in assertions, only one explicit contrast

(17) A: Last week, there was a lot of noise because Bill was building a desk. What's the noise today?

a. #My neighbor is building a DESK.

b. My NEIGHBOR is building a desk.[8]

- (17): contrastive subject focus with (apparently) obligatory deaccentuation of the VP – unlike the *wh*-questions from Experiment II
- But: is deaccentuation really obligatory?

(18) My NEIGHBOR is building a DESK.
Does this reading exist? "This time it's my NEIGHBOR making noise, and coincidentally ALSO by building a DESK."

[8] (96) from Wagner (2012)

# Example paradigm

- We need to adjust the neighbor example slightly:
- Why would someone question whether a uniquely identifiable neighbor (instead of Bill) is doing something? We need inherent contrast.
  - → In our world, the person complaining about the noise has exactly two neighbors – one to the left of their house and one to the right. Other sources of noise still exist, but the neighbors are the most salient.
- Why would someone question whether the noise is the result of building a desk again (how often do people build desks)? We need a predicate with a habitual reading.
  - → The source of the noise will be practicing the piano.

# Given object in questions

(19) A: Es ist bei mir zuhause oft total laut, weil mein linker Nachbar Musiker ist. Den ganzen Tag übt der Klavier. Jetzt ist der endlich mal im Urlaub und es ist trotzdem total laut.

*In my house, it's often very loud because my left neighbor is a musician. He practices the piano the whole day. Now he's finally on vacation for once and it's still very loud.*

B: Wie kommt's? Hat dein RECHTER Nachbar heute Klavier geübt?

B': #Wie kommt's? Hat dein RECHTER Nachbar heute KLAVIER geübt?

*How come? Did your right neighbor practice the piano today?*

# Given object in assertions

(20) A: Letzte Woche war es bei mir total laut, weil mein linker Nachbar den ganzen Tag Klavier geübt hat. Und heute ist es schon wieder total laut.
B: Wie kommt's?

*A: Last week, it was very loud at my place because my left neighbor was practicing the piano the whole day. And today it's very loud again. B: How come?*

A: Mein RECHTER Nachbar hat heute Klavier geübt.

A': Mein RECHTER Nachbar hat heute KLAVIER$_{!H*}$ geübt.

A": #Mein RECHTER Nachbar hat heute KLAVIER$_{L+H*}$ geübt.

*My right neighbor practiced the piano today.*

# Hypernym in questions

(21) A: Es ist bei mir zuhause oft total laut, weil mein linker Nachbar Musiker ist. Den ganzen Tag übt der Klavier. Jetzt ist der endlich mal im Urlaub und es war heute trotzdem total laut.

*In my house, it's often very loud because my left neighbor is a musician. He practices the piano the whole day. Now he's finally on vacation for once and it's still very loud.*

B: Wie kommt's? Hat dein RECHTER Nachbar heute ein Instrument geübt?

B': Wie kommt's? Hat dein RECHTER Nachbar heute ein INSTRUMENT geübt?

*How come? Did your right neighbor practice an instrument today?*

# Hypernym in assertions

(22) A: Letzte Woche war es bei mir total laut, weil mein linker Nachbar den ganzen Tag Klavier geübt hat. Und heute ist es schon wieder total laut.
B: Wie kommt's?

*A: Last week, it was very loud at my place because my left neighbor was practicing the piano the whole day. And today it's very loud again. B: How come?*

B: Mein RECHTER Nachbar hat heute ein Instrument geübt.

B': #Mein RECHTER Nachbar hat heute ein INSTRUMENT geübt.

*My right neighbor practiced the piano today.*

# Things to be explained

1. In assertions, *Instrument* is contrastive if accented at all, while *Klavier* needs L+H* to be contrastive.
2. In questions, an accent on *Instrument* is fine, one on *Klavier* is weird.
3. Most importantly, why is there a difference between (21B') and (22B'), i.e. a difference between the speech acts in acceptability between accented realizations of *Instrument*?

# A sketch of an analysis for assertions

In assertions, why can *Klavier* receive a small accent, while *Instrument* must be completely deaccented?

- Stipulation: L+H* is out because it would indicate that *Klavier* contrasts with *Klavier* (i.e. there is some sort of connection between L+H* and contrast in the narrow sense).
- Hypothesis: (some) other accents can highlight unexpected information without inducing contrastive focus alternatives. !H* on *Klavier*:
  - "Yes, of all the things my RIGHT neighbor could practice, it was a PIANO again" (comparatively low degree of expectedness)
- In the context of noise-generating practices one can do at home, instruments are arguably the most salient. This makes *Instrument* highly predictable, and blocks ANY accent:
  - "Yes, of all the things my RIGHT neighbor could practice, it was an INSTRUMENT again" (high degree of expectedness) $\rightarrow$ blocked
  - Any accent can then only indicate a proper, contrastive focus alternative: "an instrument, not a piano" $\rightarrow$ incoherent, underspecific

# A sketch of an analysis for questions

In questions, why can *Instrument* carry an accent, while *Klavier* must be deaccented?

- Same explanation, but in reverse: *Instrument* is highly salient and unspecific.
- This means that the question “Is it your RIGHT neighbor this time, by ALSO playing an INSTRUMENT?” is fairly reasonable; the answer is likely to be *yes*.
- “Is it your RIGHT neighbor this time, by ALSO playing a PIANO?” is quite specific; the answer is likely to be *no* or *what gave you that idea?*
- The deaccented versions, in contrast, only really question the identity of the subject:
- “So I guess someone is practicing the piano again – is it your RIGHT neighbor?”
- *Piano* can ‘sneak through’.

# The difference between the two speech acts w.r.t. to focus

- I have been unable to find any question readings where an accent automatically excludes the givenness antecedent, i.e.:

(23) *Context about left neighbor practicing the piano.*

a. My RIGHT neighbor was practicing an INSTRUMENT today.
($\rightarrow$ instead of a piano)

b. Was your RIGHT neighbor practicing an INSTRUMENT?
($\nrightarrow$ instead of a piano)

- This seems to be a fundamental difference between questions and assertions.
- (Unless I missed something, in which case I'm happy to hear from you in the discussion!)

# Focus marking in questions – speaker & addressee

- If (these particular) accents in questions do not restrict the answers available to the addressee, then what is the point of accent shift in questions?
- In assertions, accent shift can help establish co-reference, they can indicate covert focus alternatives, etc.
- All these arguably help the addressee structure their discourse model (e.g. *Dear addressee, these two expressions co-refer / despite everyone's expectations, she brought a USED bike*).
- (At least some) focus marking in questions indicates the speaker's discourse model, and only that.
- Any use that the addressee gets out of this indication is accidental.
- Let's look at one final question type to see this.

# Focus marking at the limits of questionhood

- Contrast between (24A) and (24A'): *ob*-questions do not request information from the addressee.

(24) A: I haven't heard from Peter in years.
B: Me neither.

A: #Sammelt er immer noch kubanische Zigarren?
'Does he still collect Cuban cigars?'

A': Ob er immer noch kubanische Zigarren sammelt?
*OB he still Cuban cigars collects*
'Might he still be collecting Cuban cigars, I wonder?'[9]

[9] Adapted from Truckenbrodt (2004), ex. (24)

# Focus marking at the limits of questionhood

(25) a. Ob er immer noch KUBANISCHE Zigarren sammelt?

b. Ob er immer noch kubanische Zigarren SAMMELT?

c. (He was always so into collecting stuff from Cuba.) Ob er immer noch kubanische [ZIGARREN]$_F$ sammelt?

- Since these questions are deliberately unanswerable, the highlighting of non-default referring expressions cannot constitute instructions to the addressee.
- These accents do, however, allow conclusions about which alternatives the speaker is considering.
- Since questions are non-veridical, the speaker is truly free to highlight as they wish.

# Conclusion

- With questions, there is generally more freedom to optionally indicate discourse structure (what is taken to be given, what is taken to be contrastive, etc.) because the true state of affairs is not (yet) as constrained as with assertions.
- The discourse structure that is indicated is the speaker's.
- The indication is not done for the addressee's benefit.
- Focus appears to be less contrastive in questions than in assertions (or at least not as obligatorily contrastive).

# References


[1] María Biezma. “Alternative vs Polar Questions: the cornering effect.” In: *Proceedings of SALT 19*. Ed. by Ed Cormany, Satoshi Ito, and David Lutz. 2009, pp. 37–54.

[2] Daniel Büring. “Focus, questions and givenness.” In: *Questions in Discourse*. Ed. by Malte Zimmermann, Klaus von Heusinger, and Edgar Onea. Vol. 2. Leiden: Brill, 2019, pp. 6–44.

[3] Tilman Höhle. “Über Verum-Fokus im Deutschen.” In: *Informationsstruktur und Grammatik*. Ed. by Joachim Jacobs. Opladen: Westdeutscher Verlag, 1992, pp. 139–197.

[4] Sophie Repp. “Contrast: Dissecting an elusive information-structural notion and its role in grammar.” In: *Oxford Handbook of Information Structure*. Ed. by Caroline Féry and Shinichiro Ishihara. Oxford: Oxford University Press, 2016, pp. 270–289.

[5] Roger Schwarzschild. “Givenness, AvoidF and other constraints on the placement of accent.” In: *Natural Language Semantics* 7.2 (1999), pp. 141–177.

[6] Heiko Seeliger and Sophie Repp. “Contrast and givenness in German wh-exclamatives and wh-questions.” in preparation.

[7] Heiko Seeliger and Sophie Repp. “Information-structural surprises? Contrast, givenness and (the lack of) accent shift and deaccentuation in non-assertive speech acts.” In: *Laboratory Phonology* (submitted).

[8] Yasutada Sudo. “Biased polar questions in English and Japanese.” In: *Beyond Expressives–Explorations in Use-Conditional Meaning*. Ed. by Daniel Gutzmann and Hans-Martin Gärtner. Leiden: Brill, 2013, pp. 275–296.

[9] Hubert Truckenbrodt. “Zur Strukturbedeutung von Interrogativsätzen.” In: *Linguistische Berichte* 199 (2004), pp. 313–350.

[10] Michael Wagner. “Focus and givenness: a unified approach.” In: *Contrasts and Positions in Information Structure*. Ed. by Ivona Kučerová and Ad Neeleman. Cambridge: Cambridge University Press, 2012, pp. 102–147.